

**DES PUNE UNIVERSITY**  
**School of Engineering and Technology**  
**Computer Engineering and Technology**  
**Program: B.Tech. Computer Science and Engineering**

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Roll No.: 1012411011  
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Assignment No.: 1  
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Year: Second Year  
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Title: Library Management System

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classmate  
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Java  
Assignment 1

Name: Riddhee Kulkarni  
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Aim:  
Write a program using if-else, switch-case and various loops (for, while, do-while) to solve problems. Implement a class with attributes and methods, instantiate objects and demonstrate method calls within the context of a library management system.

Theory:

1. if-else statement:  
Any if-else statement is used to make decision in a program. It checks if a condition is true or false. If it's true, the code inside the if part runs. If not, the code inside the else part runs. It's helpful when we want the program to choose between two options.  
Syntax:  

```
if (condition) {  
    // code if condition is true  
}  
else {  
    // code if condition is false  
}
```
2. for loop  
A for loop is used when we know how many times we want to repeat something. It starts with a value, checks

a condition, and then changes the value every time. It's mostly used for counting or doing something a set number of times.

Syntax:

```
for (Start; condition; update) {  
    // code to repeat  
}
```

### 3. while loop

A while loop runs the code again and again as long as the condition is true. It checks the condition first before running. If the condition is false in the beginning it won't run at all. It's useful when we don't know exactly how many times the loop should run.

Syntax:

```
while (condition) {  
    // code to repeat  
}
```

### 4. do-while loop

A do-while loop is just like a while loop, but it always runs the code at least once. This is because the condition is checked after running the code. It's helpful when we want the loop to run once before checking the condition.

Syntax:

```
do {  
    // code to repeat  
} while (condition)
```

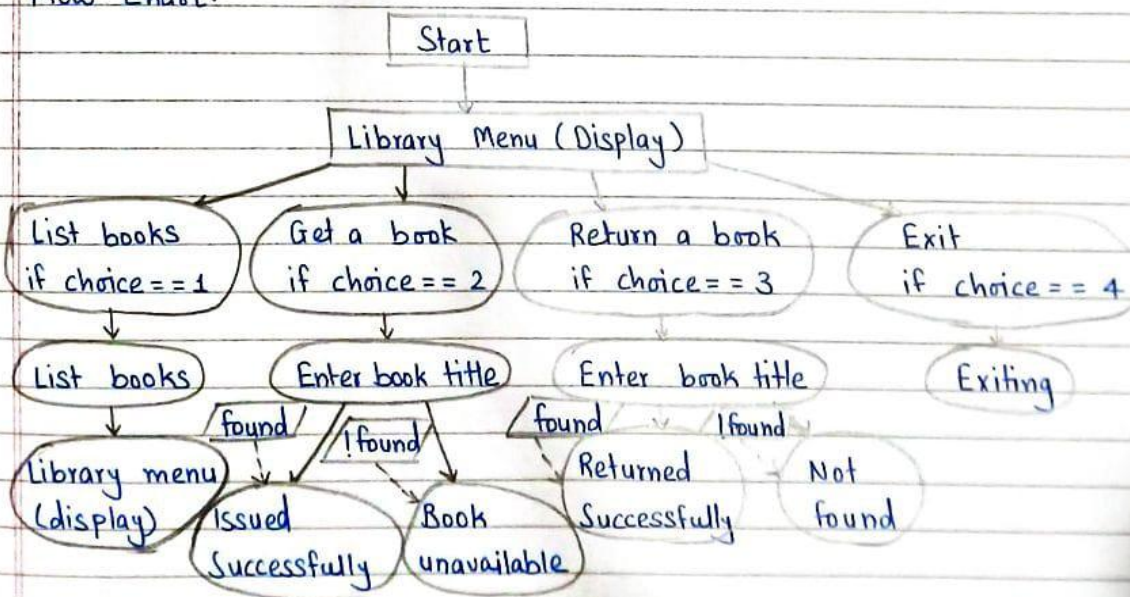
## 5 Switch-case statement

A switch statement is used when we have many fixed options and want to pick one. It checks the value and runs the matching case. If nothing matches, the default part runs. Its better than writing to many if else statements.

Syntax:

```
switch (value) {  
    case option1  
        // code  
        break;  
    case option2  
        //code  
        break;  
    default:  
        // code if no match  
}
```

Flow chart:





Code:

```
import java.util.Scanner;

class Book {
    String title;
    String author;
    int quantity;

    Book(String title, String author, int quantity) {
        this.title = title;
        this.author = author;
        this.quantity = quantity;
    }
}

public class LibrarySystem
{
    public Book[] books =
    {
        new Book("The Alchemist", "Paulo Coelho", 3),
        new Book("To Kill a Mockingbird", "Harper Lee", 2),
        new Book("1984", "George Orwell", 1),
        new Book("The Great Gatsby", "F. Scott Fitzgerald", 1),
        new Book("Sapiens", "Yuval Noah Harari", 2)
    };

    public void listBooks()
    {
        System.out.println("\nAvailable Books:");
        for (int i = 0; i < books.length; i++)
        {
            if (books[i].quantity > 0)
            {
                System.out.println("- " + books[i].title + " by " + books[i].author);
            }
        }
    }

    public void getBook(String title)
    {
        boolean found = false;
        for (int i = 0; i < books.length; i++)
        {
            if (books[i].title.equals(title))
            {
                found = true;
                if (books[i].quantity > 0)
                {
                    books[i].quantity--;
                    System.out.println("Book \"" + books[i].title + "\" issued successfully.");
                }
            }
        }
    }
}
```

```
        else
        {
            System.out.println("No copies available right now.");
        }
        break;
    }
}
if (!found)
{
    System.out.println("Book not found.");
}
}

public void returnBook(String title)
{
    boolean found = false;
    for (int i = 0; i < books.length; i++)
    {
        if (books[i].title.equals(title))
        {
            books[i].quantity++;
            found = true;
            System.out.println("Book \"\" + books[i].title + "\" returned successfully.");
            if(books[i].quantity > 0)
            {
                System.out.println("Book is now available");
            }
        }
    }
}
```

```
        break;
    }
}
if (!found)
{
    System.out.println("Book not found in system.");
}
}

public static void main(String[] args)
{
    Scanner sc = new Scanner(System.in);
    LibrarySystem library = new LibrarySystem();
    int choice;

    do {
        System.out.println("\n1. List books");
        System.out.println("2. Get a book");
        System.out.println("3. Return a book");
        System.out.println("4. Exit");
        System.out.print("Enter your choice: ");
        choice = sc.nextInt();
        sc.nextLine();

        if (choice == 1)
```

```
        library.listBooks();
    }
    else if (choice == 2)
    {
        System.out.print("Enter book title to get: ");
        String getTitle = sc.nextLine();
        library.getBook(getTitle);
    }
    else if (choice == 3)
    {
        System.out.print("Enter book title to return: ");
        String returnTitle = sc.nextLine();
        library.returnBook(returnTitle);
    }
    else if (choice == 4)
    {
        System.out.println("Exiting");
    }
    else
    {
        System.out.println("Invalid choice");
    }
} while (choice != 4);

sc.close();
}
```

Output:

```
1. List books
2. Get a book
3. Return a book
4. Exit
Enter your choice: 1

Available Books:
- The Alchemist by Paulo Coelho (Copies: 3)
- To Kill a Mockingbird by Harper Lee (Copies: 2)
- 1984 by George Orwell (Copies: 1)
- The Great Gatsby by F. Scott Fitzgerald (Copies: 1)
- Sapiens by Yuval Noah Harari (Copies: 2)

1. List books
2. Get a book
3. Return a book
4. Exit
Enter your choice: 2
Enter book title to get: 1984
Book "1984" issued successfully.

1. List books
2. Get a book
3. Return a book
4. Exit
Enter your choice: 2
Enter book title to get: blahblah
Book not found.
```

```
1. List books
2. Get a book
3. Return a book
4. Exit
Enter your choice: 3
Enter book title to return: blahblah
Book not found in system.

1. List books
2. Get a book
3. Return a book
4. Exit
Enter your choice: 3
Enter book title to return: 1984
Book "1984" returned successfully.
Book is now available

1. List books
2. Get a book
3. Return a book
4. Exit
Enter your choice: 4
Exiting
```

• Conclusion:

Through this program we get to learn how to implement if-else statement, loops (for, while, do-while) and switch cases.